

Project Report

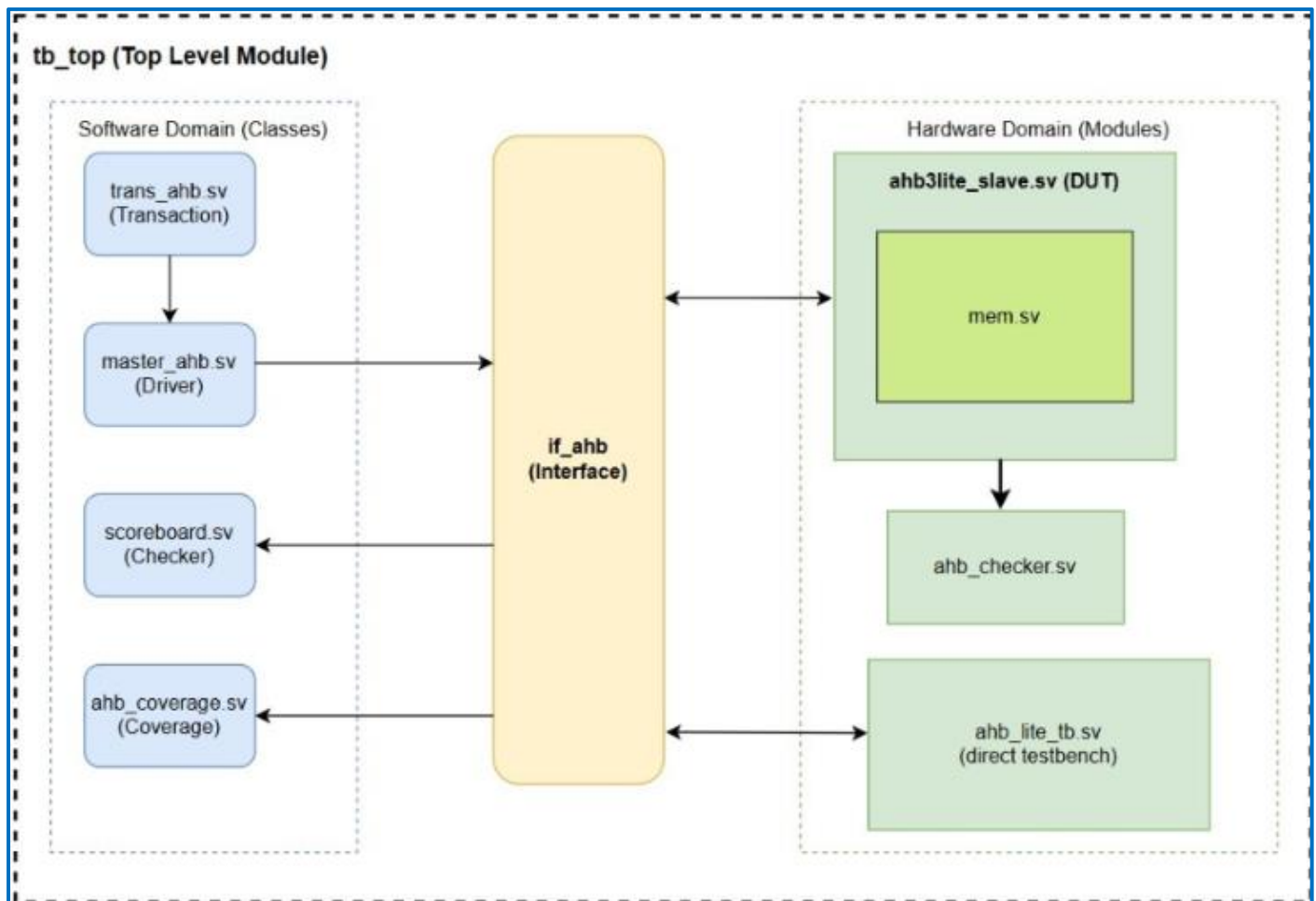


Course Title	EE5214: Verification of Digital Systems
Project Title	Functional and Formal Verification of AMBA AHB-Lite RAM
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1. Testbench Architecture

Testbench architecture is presented in the figure below.



2. Functional Verification Summary

2.1 Overall Test Plan Status

The AHB-Lite test plan contains a total of **10 distinct test scenarios** designed to verify the correct operation of the **AMBA AHB-Lite Slave (RAM) verification**. As of the latest execution cycle, **100% (10 out of 10)** of the tests have been **successfully completed**.

2.2 Scope of Testing

The testing matrix covered fundamental and advanced AMBA AHB-Lite protocol operations across simple transfers, varying burst types, and data sizes. The following different types of test scenarios have been targeted and verified in functional and formal verification of AMBA AHB-Lite Slave RAM.

- **SINGLE Transfers (4 tests):** Verified base-level write and read operations across different data widths (Byte, Halfword, and Word). This also included a test specifically targeting back-to-back pipelined single transfers with no IDLE cycles.
- **INCR Bursts (4 Tests):** Verified incrementing burst operations (INCR4, INCR8, INCR16) across word-sized data boundaries. This included verifying behavior when HREADY stall conditions are introduced.

- **WRAP Bursts (2 Tests):** Verified wrapping burst logic (WRAP4 and WRAP8) to ensure the address calculation safely wraps memory boundaries according to the HSIZE and HBURST protocol rules.

2.3 Test Scenario Breakdown and Status

The details of all the test cases executed in this project are given in the table below.

Transfer Type	Description	Status
SIMPLE	Single Word Write/Read	COMPLETED
	Single Halfword Write/Read	COMPLETED
	Single Bytes Write/Read	COMPLETED
	Back-to-Back Write/Read	COMPLETED
BURST	INCR4 Word Write/Read	COMPLETED
	INCR8 Word Write/Read	COMPLETED
	INCR16 Word Write/Read	COMPLETED
	WRAP4 Word Write/Read	COMPLETED
	WRAP8 Word Write/Read	COMPLETED
	INCR4 with Wait States	COMPLETED

3. Functional Coverage

3.1 Summary

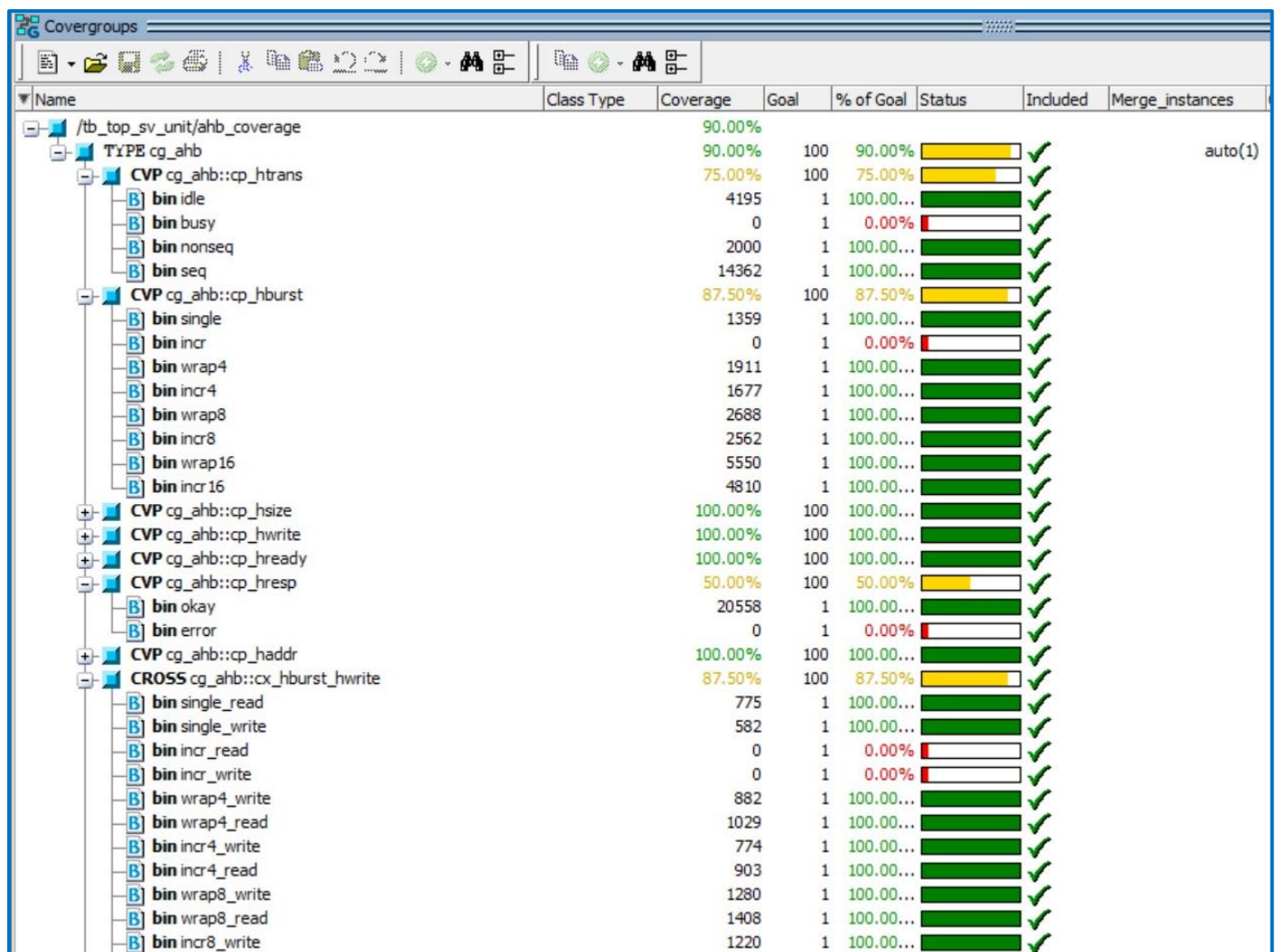
A summary of functional coverage is given in the table below.

Coverage Type	Coverpoint/Cross	Description/Goal	Bins/Scenerios
Transfer Type	Cp_htrans	Tracks the type of current AHB transfer	IDLE, BUSY, NONSEQ, and SEQ
Burst Type	Cp_hburst	Monitors all standard AHB-Lite burst protocols	SINGLE, INCR, WRAP4/8/16, and INCR4/8/16
Data Size	Cp_hsize	Tracks the width of the data payload	Byte, Halfword, and Word (32-bit). Unsupported sizes are ignored
Transfer Direction	Cp_hwrite	Monitors Read vs. Write operations	Read (0) and Write (1)
Slave Status	Cp_hready	Tracks if the slave is ready or inserting wait states	Ready (1) and Stalled (0)
Response	Cp_hresp	Ensures the slave responds correctly ¹	OKAY response
Address Space	Cp_haddr	Verifies distribution across the address map	Four equal memory regions (0x0000 to 0xFFFF)
Burst Direction	Cx_hburst_hwrite	Cross: Validates every burst type for both Read/Write	e.g., WRAP8 Write, INCR16 Read, etc.

Size Direction	Cx_size_write	Cross: Ensures all data widths work for both Read/Write	e.g., Byte Write, Word Read, etc.
Wait States	Cx_htrans_hready	Cross: Track if the design handles wait states during active sequential bursts	Specifically monitors sequential transfers stalled by HREADY = 0
Response	Cp_hresp	Tracks slave response	OKAY and ERROR

3.2 Coverage Results

Code and functional coverage analysis have been performed on Questasim. The results are presented in the figure below.



3.3 Coverage Results Summary

A coverage results summary is presented in the table below.

Covergroup Element	Coverage	Detailed Analysis & Results
Total Coverage	90 %	High, but limited by specific missing protocol states
Address Map (cp_haddr)	100 %	Fully Covered: All four address quadrants were hit uniformly.
Data Size (cp_hsize)	100%	Fully Covered: Byte, Halfword, and Word transfers were all verified.
Ready/Stall (cp_hready)	100 %	Fully Covered: Captured 1,000 instances of HREADY=0, proving the system handles wait states correctly.
Burst Type (cp_hburst)	87.50 %	GAP: All fixed-length bursts (SINGLE, WRAP, INCR4/8/16) are covered. The INCR (undefined length) burst was never exercised.
Transfer Type (cp_htrans)	75 %	GAP: IDLE, NONSEQ, and SEQ are fully covered. However, BUSY bin has 0 hits, indicating the master never inserted wait states during a burst.
Read/Write (cp_hwrite)	100 %	Fully Covered: Both directions are thoroughly tested.
Slave Ready (cp_hready)	100 %	Fully Covered: The design correctly handles slave wait states.
Response (cp_hresp)	50 %	GAP: Only OKAY responses were received (20,558 hits). No ERROR responses were triggered during simulation.
Cross Coverage	87.50 % - 100 %	GAP: Missing coverage for incr_read and incr_write because the INCR burst type was not exercised.

4. Formal Verification

In the Formal Verification of the design, we have performed comprehensive verification of the RoaLogic AHB3-Lite SRAM Controller using Formal Verification in JasperGold. We have written about 15 SystemVerilog Assertions (SVA) based on the AMBA AHB3-Lite Specifications. Developed a dedicated Assumption Module to constraint the Formal Master to legal behavior. Used the bind methodology to connect the checker to the RTL without modifying the source code.

4.1 Assertions Summary

The summary of assertions is given in the table below.

ID	Specification	Property Description	AHB Section
SPEC-1	HADDR Alignment	Address must be aligned to HSIZE (e.g., 32-bit word must end in 2'b00)	3.4
SPEC-2	HTRANS Sequencing	SEQ must always be preceded by a NONSEQ or another SEQ	3.2
SPEC-3	Wait State Stability	Master must keep Address/Control signals stable when HREADY is LOW	3.6
SPEC-4	Burst Continuity	Fixed-length bursts (INCR4) must complete all beats without interruption	3.5
SPEC-5	1KB Boundary	Incrementing bursts must not cross the 1KB address limit	3.5
SPEC-6	\ ERROR Response	Two-cycle error sequence: Cycle 1 (HREADY = 0), Cycle 2 (HREADY = 1)	5.1
SPEC-7	Illegal BUSY	BUSY transfer type is prohibited during a SINGLE burst	3.5.1
SPEC-8	Reset State	HREADYOUT must be HIGH and HRESP must be OKAY immediately after reset	7.1
SPEC-9	Write Validation	A valid write command (HSEL + HWRITE) should not be rejected	7.1
SPEC-10	HSIZE Protection	HSIZE must not exceed the physical data bus width (32-bit)	10.1
SPEC-11	Liveness Check	Slave must eventually assert HREADYOUT (No deadlock)	Functional
SPEC-12	Busy Termination	Fixed bursts cannot be terminated by deselecting Slave during BUSY	3.5.1
SPEC-13	Busy Response	Slave must provide a Zero Wait-State OKAY response to BUSY cycles	3.5.1
SPEC-14	Address Increment	Verifies HADDR increments correctly based on HSIZE during SEQ	3.2
SPEC-15	WRAP4 Boundary	Verifies 4-beat wrapping bursts stay within the 16-byte boundary	3.5

4.2 Assumptions Summary

The summary of assertions is given in the table below.

ID	Assumption Name	Description	Protocol Constraint
ASMP-1	Wait Stability	Master must keep Address and Write signals stable if HREADY is LOW	Stability (Sec 3.6)
ASMP-2	Address Alignment	Constraints HADDR to be aligned based on HSIZE (Halfword/Word)	Alignment (Sec 3.4)
ASMP-3	1KB Boundary	Ensures incrementing bursts never cross the architectural 1KB limit	Bursting (Sec 3.5)
ASMP-4	HTRANS Sequencing	SEQ transfers must follow a valid NONSEQ or SEQ start	Sequencing (Sec 3.2)
ASMP-5	Single vs SEQ	Prohibits SEQ transfers when the burst type is set to SINGLE	Burst Rules
ASMP-6	INCR4 Completion	Forces the Formal tool to finish 4-beat bursts once started	Burst Integrity
ASMP-7	SEQ Address Calculation	Constraints the Master to increment address correctly during SEQ beats	Address Logic
ASMP-8	Legal HSIZE	Restricts HSIZE to Word length (32-bit) or less for this specific Slave	Data Width
ASMP-9	Full Master Stability	Extends stability to HTRANS, HSIZE, and HBURST during wait states	Stability (Sec 3.6)
ASMP-10	WRAP4 Boundary	Constraints the address wrapping math for 4-beat bursts	Wrapping Logic
ASMP-11	Single vs. BUSY	Prohibits the use of BUSY cycles during SINGLE transfers	Sec 3.5.1
ASMP-12	WRAP Alignment	Ensures WRAP4 address math stays in boundary	Wrapping Logic
ASMP-13	HSEL Liveness	Keeps the Slave selected to focus analysis on active cycles	Verification Focus
ASMP-14	HREADY Liveness	Ensures HREADY eventually returns HIGH (prevents infinite wait stalls)	Progress Guarantee

4.3 Jasper Gold Results Summary

The formal verification has achieved the following results:

- **Functional Coverage:** 90.625%
- **Property Proving:** 73.3% (11 of 15 assertions mathematically proven)
- **Bugs Identified:** 2 Critical Protocol Violations discovered in the RoaLogic DUT

4.4 Property Table (JasperGold)

Property Table							
		No filter	Filter on name				
Properties	Type	Name	Engine	Bound	Traces	Time	
✓	Assert	ahb3liten.checker_inst.assert_haddr_alignment	Hp (1)	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_haddr_alignment:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_htrans_sequencing	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_htrans_sequencing:precondition1	Hp	1	1		
✗	Assert	ahb3liten.checker_inst.assert_wait_state_stability	Hp	2	1		
✓	Cover (related)	ahb3liten.checker_inst.assert_wait_state_stability:precondition1	Hp	1	1		
✗	Assert	ahb3liten.checker_inst.assert_burst_continuity_incr4	Hp	2	1		
✓	Cover (related)	ahb3liten.checker_inst.assert_burst_continuity_incr4:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_1kb_boundary_cross	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_1kb_boundary_cross:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_error_resp_timing	PRE	Infinite	0		
✗	Cover (related)	ahb3liten.checker_inst.assert_error_resp_timing:precondition1	PRE	Infinite	0		

Property Table							
		No filter	Filter on name				
Properties	Type	Name	Engine	Bound	Traces	Time	
✓	Assert	ahb3liten.checker_inst.assert_no_busy_in_single	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_no_busy_in_single:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_reset_behavior	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_reset_behavior:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_write_condition	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_write_condition:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_valid_hsize	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_valid_hsize:precondition1	Hp	1	1		
✓	Assert (live)	ahb3liten.checker_inst.assert_liveness_hready	N (14)	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_liveness_hready:precondition1	Hp	1	1		
●	Assume	ahb3liten.checker_inst.assume_master_wrap_align	?		0		
✓	Cover (related)	ahb3liten.checker_inst.assume_master_wrap_align:precondition1	Hp	1	1		

Total: 63 Filtered: 63 Selected: 1 Validity: 40:7:0:16 Run: 16:0:0:4

Property Table							
		No filter	Filter on name				
Properties	Type	Name	Engine	Bound	Traces	Time	
✓	Assert	ahb3liten.checker_inst.assert_no_busy_term	PRE	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_no_busy_term:precondition1	Hp	1	1		
✗	Assert	ahb3liten.checker_inst.assert_busy_ok_response	Hp	2	1		
✓	Cover (related)	ahb3liten.checker_inst.assert_busy_ok_response:precondition1	Hp	1	1		
✓	Assert	ahb3liten.checker_inst.assert_seq_addr_calc	Hp (1)	Infinite	0		
✓	Cover (related)	ahb3liten.checker_inst.assert_seq_addr_calc:precondition1	Hp	1	1		
✗	Assert	ahb3liten.checker_inst.assert_wrap4_boundary	Hp	1	1		
✓	Cover (related)	ahb3liten.checker_inst.assert_wrap4_boundary:precondition1	Hp	1	1		
✓	Cover	ahb3liten.checker_inst.cover_full_incr16	Hp	15	1		
✓	Cover	ahb3liten.checker_inst.cover_wrap4_event	Hp	3	1		
✗	Cover	ahb3liten.checker_inst.cover_error_response	PRE	Infinite	0		
●	Assume	ahb3liten.asm_inst.asm_reset_idle	?		0		

Total: 63 Filtered: 63 Selected: 1 Validity: 40:7:0:16 Run: 16:0:0:4

Property Table							
No filter		Filter on name		a.b P			
Type	Name	Engine	Bound	Traces	Time		
Assume	ahb3liten.asm_inst.asm_reset_idle	?		0			
Cover (related)	ahb3liten.asm_inst.asm_reset_idle:precondition1	PRE	Infinite	0			
Assume	ahb3liten.asm_inst.asm_wait_stability	?		0			
Cover (related)	ahb3liten.asm_inst.asm_wait_stability:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_addr_alignment	?		0			
Cover (related)	ahb3liten.asm_inst.asm_addr_alignment:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_1kb_boundary	?		0			
Cover (related)	ahb3liten.asm_inst.asm_1kb_boundary:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_htrans_seq_rule	?		0			
Cover (related)	ahb3liten.asm_inst.asm_htrans_seq_rule:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_single_no_seq	?		0			
Cover (related)	ahb3liten.asm_inst.asm_single_no_seq:precondition1	Hp	1	1			
Total: 63		Filtered: 63		Selected: 1		Validity: 40:7:0:16 Run: 16:0:0:4	

Property Table							
No filter		Filter on name		a.b P			
Type	Name	Engine	Bound	Traces	Time		
Assume	ahb3liten.asm_inst.asm_legal_hsize	?		0			
Cover (related)	ahb3liten.asm_inst.asm_legal_hsize:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_wait_stable	?		0			
Cover (related)	ahb3liten.asm_inst.asm_wait_stable:precondition1	Hp	2	1			
Assume	ahb3liten.asm_inst.asm_wrap4_math	?		0			
Cover (related)	ahb3liten.asm_inst.asm_wrap4_math:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_single_no_busy	?		0			
Cover (related)	ahb3liten.asm_inst.asm_single_no_busy:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_master_wrap_align	?		0			
Cover (related)	ahb3liten.asm_inst.asm_master_wrap_align:precondition1	Hp	1	1			
Assume	ahb3liten.asm_inst.asm_hsel_liveness	?		0			
Assume (live)	ahb3liten.asm_inst.asm_hready_loop	?		0			
Total: 63		Filtered: 63		Selected: 1		Validity: 40:7:0:16 Run: 16:0:0:4	

4.5 Analysis

The properties were analyzed and categorized based on their JasperGold status (Proven, CEX, or Unreachable).

Category	Status	Count	Description
Protocol Foundations	Proven	11	Core rules such as Address Alignment, Reset Behavior, and Sequencing
Design Bugs	CEX	2	Identified failures where the DUT violated AMBA 3 Spec
Environmental Gaps	CEX	2	Scenarios where the Formal Master was under-constrained, allowing it to perform illegal signal “wiggles”.
Exclusions	Unreachable	3	Scenarios physically impossible in this specific SRAM design

4.6 Detailed Bug Report (Counter-Example Analysis)

Two significant design flaws were isolated in the provided DUT. These represent actual failures of the RTL to meet the AMBA AHB3-Lite Specification.

Bug#1: Illegal Wait-State during BUSY Cycle (assert_busy_ok_response)

- **Spec Reference (Section 3.5.1):** Slaves must provide a zero wait-state OKAY response to BUSY.
- **Failure:** The DUT incorrectly de-asserted HREADYOUT (Wait state) while the Master was in a BUSY state.
- **Impact:** This could cause the AHB bus to stall unnecessarily, decreasing system throughput.

Bug#2: Wrapping Address Calculation Error (assert_wrap4_boundary)

- **Spec Reference (Section 3.5.2):** Address calculation for wrapping bursts.
- **Failure:** During a 4-beat wrapping word transfer, the DUT's internal address incrementor failed to perform the correct modulo-16 wrap-around.
- **Impact:** Data corruption occurs when crossing the wrapping boundary.

4.7 Coverage Hole & Vacuity Analysis

The final functional coverage was recorded at 90.625%. The remaining 9.375% of “unreachable” coverage was analyzed for vacuity:

1. **ERROR Response (cover_error_response):** Unreachable because this specific SRAM is an “OKAY-only” responder.
2. **1KB Boundary:** Unreachable due to the MEM_DEPTH constraints in the formal setup, which prevented the tool from exploring the full 32-bit address space.

4.8 Verification Environment Integrity

As shown in the JasperGold Property Table, all assumptions were active (Blue Circle) and their preconditions were covered (Green Check), proving that the results are not “vacuous” and that the Master was successfully stimulated to perform legal AHB transfers.

4.9 Waveform Analysis of assert_busy_ok_response

Finding: Critical Protocol Violation (**Design Bug**)

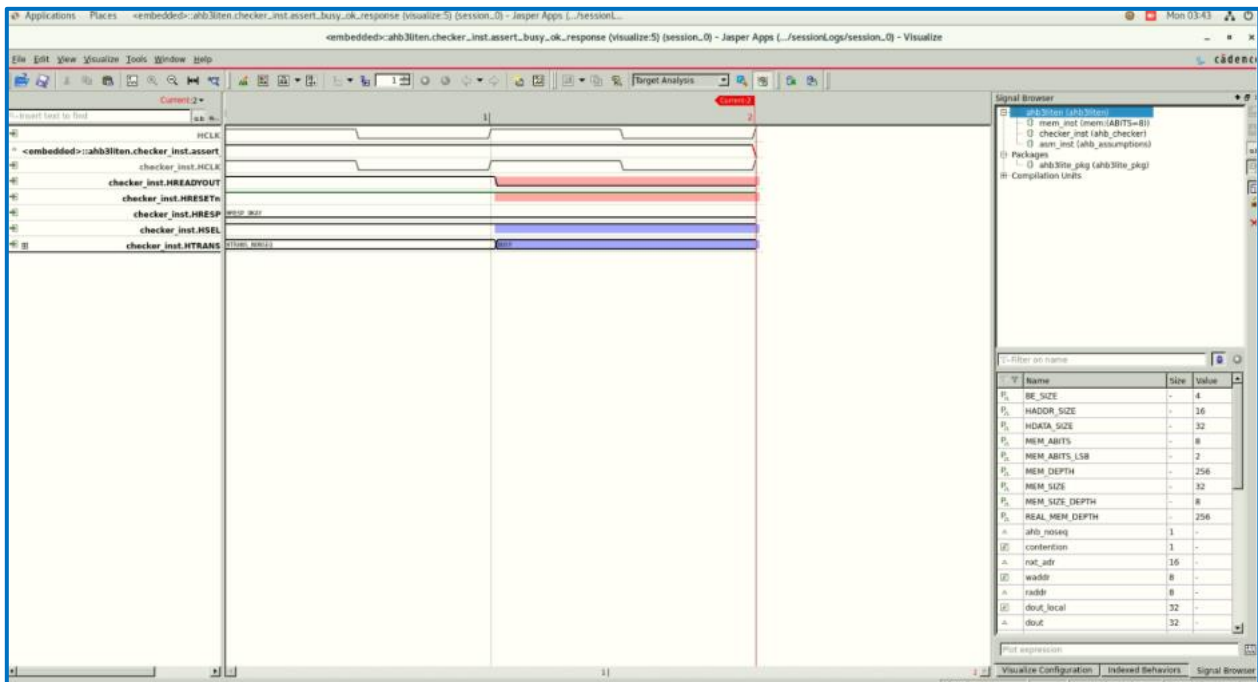
AHB3-Lite Spec Reference (Section 3.5.1)

Slaves must provide a zero wait-state OKAY response to BUSY transfers. This means HREADYOUT must remain high, and HRESP must remain OKAY whenever the Master is in a BUSY state.

Waveform Observations

In the figure given below, we observe a counter-example (CEX) because the Slave (DUT) fails to handle a BUSY transfer correctly.

- **Cycle 1:** The Master drives HTRANS to BUSY (2'b01).
- **The Violation:** At the same moment, the Slave de-asserts HREADYOUT (pulling it to 0).
- **Impact:** The Slave is incorrectly inserting a WAIT state during a cycle it is supposed to ignore.



4.10 Waveform Analysis of assert_wait_state_stability

Finding: Environmental Gap (Illegal Master Behavior)

AHB3-Lite Spec Reference (Section 3.6)

During a wait state (when HREADY is low), the Master must keep the address and all control signals stable. They can only change in the cycle after HREADY is sampled high.

Waveform Observations

In the figure given below, counter-example generated by JasperGold for the `assert_wait_state_stability` property is presented.

- **Cycle 1:** The Slave (DUT) pulls HREADY low to insert a wait state.
- **Cycle 2:** While HREADY is still low, the Master changes the HADDR, HSIZE, and HBURST signals.
- **Cycle 3:** The assertion triggers a failure because it detected a change in control signals during an active wait state.

